Spacecraft ACDS Homework 1

Fausto Vega

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1 Mission Description and Requirements

This report analyzes the Starling mission. It consists of four 6-unit CubeSats flying in a sun synchronous circular orbit at an altitude of 550 km. The high-level mission objectives consist of swarm maneuver planning and execution, relative navigation, and autonomous coordination between the spacecraft [1]. Onboard the satellites, there are two commercial star-trackers, dual-band GPS receivers, reaction wheels, a Globalstar beacon radio, and an S-band radio [2]. The actuation includes a cold gas propulsion system called Hamlet providing up to 30 m/s of delta-V for maneuvers [3]. The satellite is a Blue Canyon XB 6U cubesat which has a the pointing accuracy is 8 arcseconds degrees, pointing stability of 7.2 arcseconds, slew rate of 360 degrees per minute, and pointing stability of 1 arcsecond/second [4].

2 Spacecraft Model

Part 1: We assume the dimensions of the 6U cubesat are 20 cm x 10 cm x 34 cm. Another assumption is the size of the panels are 30 cm x 100 cm (thickness not taken into account). The body axes are defined in Figure 1. The origin is defined to be the center of the spacecraft bus.
Part 2: We model the spacecraft as 3 rectangular prisms. One larger rectangular prism for the spacecraft body and another 2 thin rectangular prisms representing the solar panels. This decomposition along with the dimensions of each rectangular prism are depicted in Figure 2. To find the center of mass, we apply Equation 1, where m_k is the mass of each of the rectangular prisms and x_k is the distance from the body frame origin to the center of mass of each rectangular prism. We assume the mass of the satellite bus m_1 is 12 kg, the mass of each solar panel (m_2, m_3) is 1 kg. Another assumption is that the solar panels are at 45 degrees from the body X-axis and Y-axis.

$$X_{com} = \frac{\sum_{k=1}^{n} m_k x_k}{\sum_{k=1}^{n} m_k} \tag{1}$$

COM Calculation

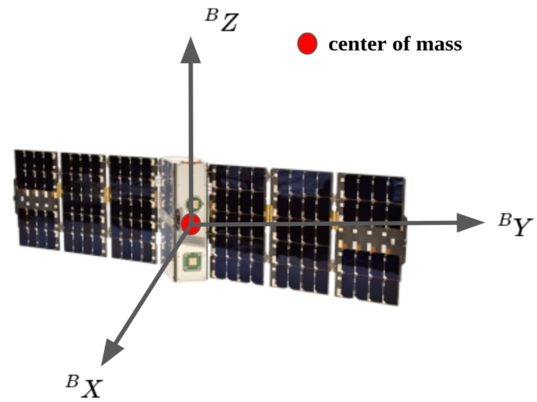
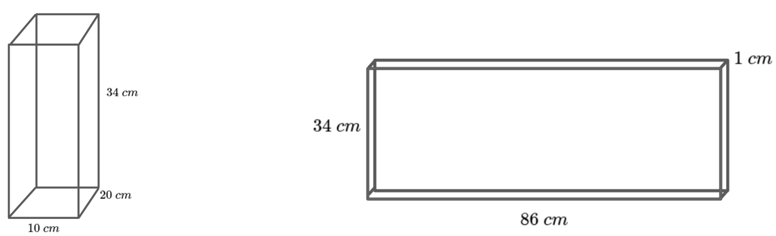


Figure 1: Body axes definition for Starling mission spacecraft



(a) Starling bus rectangle decomposition and dimensions

(b) Starling solar panels rectangular decomposition and dimensions

Figure 2: Starling 6U Cubesat Rectangular Decomposition and Dimensions

$m_1 = 12$ kg, $r_1 = (0, 0, 0)$ cm
$m_2 = 1$ kg, $r_2 = (35.36, 35.36, 0)$ cm
$m_3 = 1$ kg, $r_3 = (-35.36, -35.36, 0)$ cm
$r_{\text{com}} = [0, 0, 0]$ cm
This makes sense spacecraft because the spacecraft is symmetric.
To calculate the body-frame moment of inertia, we need to combine the individual inertias of each rectangular prism. We do this by adding the inertias of each individual term and adding the point-mass terms about the combined centered of mass (Equation 2).

$$J_{\text{combined}} = J_1 + J_2 + J_3 + m_1(r_1^T r_1 I - r_1 r_1^T) + m_2(r_2^T r_2 I - r_2 r_2^T) + m_3(r_3^T r_3 I - r_3 r_3^T) \quad (2)$$

In this equation, the term r_k is the distance from each rectangular prism center of mass to the center of mass of the spacecraft. Since the COM of the spacecraft is the origin of the body frame, the r terms remain the same from the previous part. $^{B}J = \begin{bmatrix} 5915.99 & -2500.66 & 0.0 \\ -2500.66 & 3949.49 & 0.0 \\ 0.0 & 0.0 & 7168.15 \end{bmatrix} \text{kg} \cdot \text{cm}^2$

Part 3: The principle axes of a spacecraft is the basis where the body frame moment of inertia is diagonal. We calculate by taking an eigen decomposition of the body frame moment of inertia. The set of eigenvectors becomes the principle axes P.

$P = \begin{bmatrix} 0.56306 & 0.0 & 0.826416 \\ 0.826416 & 0.0 & -0.56306 \\ 0.0 & 1.0 & 0.0 \end{bmatrix}$

The rotation between the principle frame and body frame is defined by matrix P. To transform the body frame inertia into the principle frame inertia, we apply the following: $^{P}J = P^{T\,P}JP$.

Part 4: We can discretize the satellite bus into 6 faces, and the solar panels into 5 faces each. This is a total of 16 faces to represent the spacecraft. The next section will specify the centroid, normal vector, and area of each piece.

Spacecraft bus:

Face 1:
Centroid: $[0, 5, 0]$ cm
Normal Vector: $[0, 1, 0]$
Area: 680 cm^2

Face 2:
Centroid: $[0, -5, 0]$ cm
Normal Vector: $[0, -1, 0]$
Area: 680 cm^2

Face 3:
Centroid: $[10, 0, 0]$ cm
Normal Vector: $[1, 0, 0]$
Area: 340 cm^2

Face 4:
Centroid: $[-10, 0, 0]$ cm
Normal Vector: $[-1, 0, 0]$
Area: 340 cm^2

Face 5:
Centroid: $[0, 0, 17]$ cm
Normal Vector: $[0, 0, 1]$
Area: 200 cm^2

Face 6:
Centroid: $[0, 0, -17]$ cm
Normal Vector: $[0, 0, -1]$
Area: 200 cm^2

Solar Panel 1:

Face 1:
Centroid: $[35.36, 35.36, 0]$ cm
Normal Vector: $[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$
Area: 2924 cm^2

Face 2:
Centroid: $[35.36, 35.36, 0]$ cm
Normal Vector: $[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$
Area: 2924 cm^2
Face 3:
Centroid: [35.36, 35.36, 17.0]
Normal Vector: $[0, 0, 1]$
Area: 86 cm^2
Face 4:
Centroid: [35.36, 35.36, -17.0]
Normal Vector: [0, 0, -1]
Area: 86 cm^2
Face 5:
Centroid: [70.71, 70.71, 0.0]
Normal Vector: $[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$
Area: 86 cm^2
Solar Panel 2:
Face 1:
Centroid: $[-35.36, -35.36, 0]$ cm
Normal Vector: $[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$
Area: 2924 cm^2
Face 2:
Centroid: $[-35.36, -35.36, 0]$ cm
Normal Vector: $[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$
Area: 2924 cm^2
Face 3:
Centroid: [-35.36, -35.36, 17.0]
Normal Vector: $[0, 0, 1]$
Area: 86 cm^2
Face 4:
Centroid: [-35.36, -35.36, -17.0]
Normal Vector: [0, 0, -1]
Area: 86 cm^2
Face 5:
Centroid: [-70.71, -70.71, 0.0]
Normal Vector: $[-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$
Area: 86 cm^2

3 Dynamics

Equation 3 calculates the gravitational acceleration, and Equation 4 calculates the J2 acceleration. The variable r is the position and v is the velocity. For now, we assume that the total acceleration is the sum of both of these accelerations.

$$a_g = \frac{-\mu}{||r||^3} r \tag{3}$$

$$a_{J_2} = -\frac{3}{2}\frac{J_2 \mu R_E^2}{r^5}\begin{bmatrix}\left(1-5\left(\frac{x_3}{r}\right)^2\right)x_1\\ \left(1-5\left(\frac{x_3}{r}\right)^2\right)x_2\\ \left(3-5\left(\frac{x_3}{r}\right)^2\right)x_3\end{bmatrix} \tag{4}$$

$$a_{\text{total}} = a_g + a_{J2} \tag{5}$$

We implemented an RK4 integrator and simulated the dynamics for 10 revolutions. The resulting trajectory is in Figure 3.

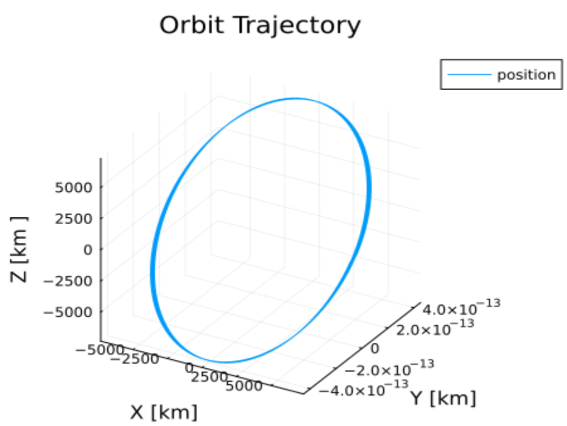


Figure 3: Orbit Trajectory

4 Euler's Equation

Part 1: The ODE's for Euler's equation along the principle axes are in Equation 6.

$$\begin{aligned} {}^PJ_{11}{}^P\dot{\omega}_1 + ({}^PJ_{33} - {}^PJ_{22}){}^P\omega_2{}^P\omega_3 &= {}^P\tau_1 \\ {}^PJ_{22}{}^P\dot{\omega}_2 + ({}^PJ_{11} - {}^PJ_{33}){}^P\omega_1{}^P\omega_3 &= {}^P\tau_2 \\ {}^PJ_{33}{}^P\dot{\omega}_3 + ({}^PJ_{22} - {}^PJ_{11}){}^P\omega_1{}^P\omega_2 &= {}^P\tau_3 \end{aligned} \tag{6}$$

Part 2: The initial condition simulated had a magnitude of 10 RPM. The perturbation had a magnitude of 0.1 $\frac{\text{rad}}{\text{s}}$ and the direction was sampled from a normal distribution. Figure 4 shows a spin around the minor axis that is marginally stable. There is nutation around the y and z axes of the principle frame. Figure 6 shows a spin around the major axis that is also marginally stable. There is nutation around the x and z principle axes. Finally, Figure 5 shows an unstable spin around the intermediate axis. We can observe the switching behavior that occurs in the x and y-directions of the angular velocity from positive to negative.

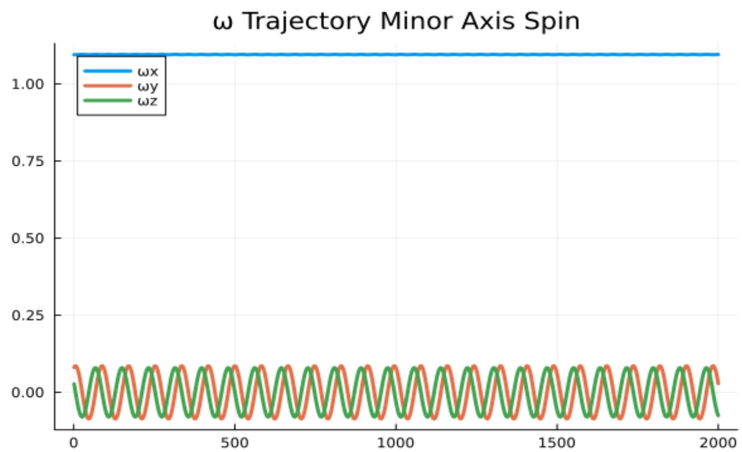


Figure 4: Minor Axis Spin Angular Velocity Trajectory

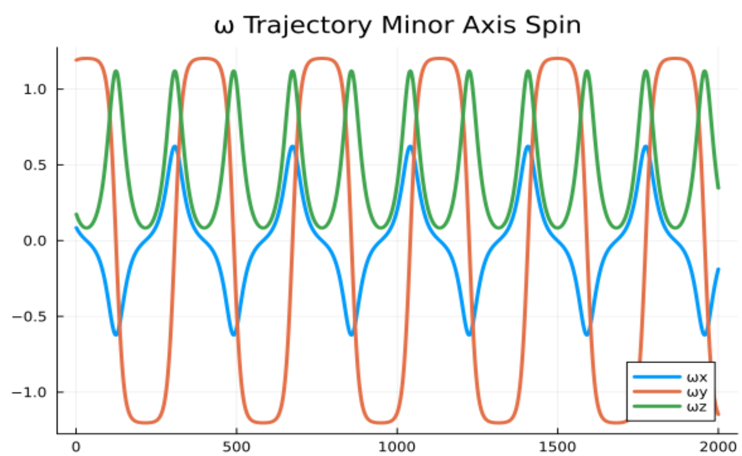


Figure 5: Intermediate Axis Spin Angular Velocity Trajectory

Part 3: Figures 7 and 8 show a plot of the momentum sphere with the equilibrium points. The green markers indicate the stable equilibrium points

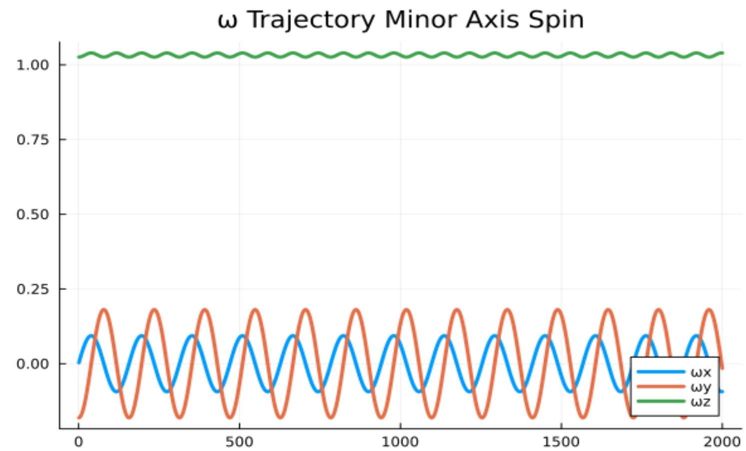


Figure 6: Major Axis Spin Angular Velocity Trajectory

(minor/major axis) and the red markers are the unstable equilibria (intermediate axis).

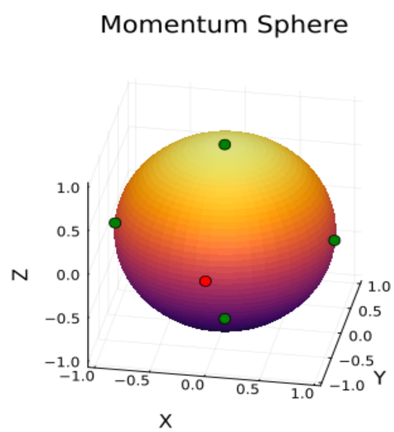


Figure 7: Momentum Sphere View 1

Next, we show several momentum trajectories on the momentum sphere from spinning about each of the principle axes. The momentum trajectories are depicted by the blue lines while the equilibrium points are represented by markers. The red marker indicates unstable equilibria (along the intermediate axis), while green are the stable equilibria. For the intermediate axis, the initial condition has a magnitude of 20 RPM with a perturbation in a random direction with a magnitude of 0.1 $\frac{rad}{s}$. The momentum trajectory about the intermediate axis is shown in Figure 9. It shows the same switching behavior we observed

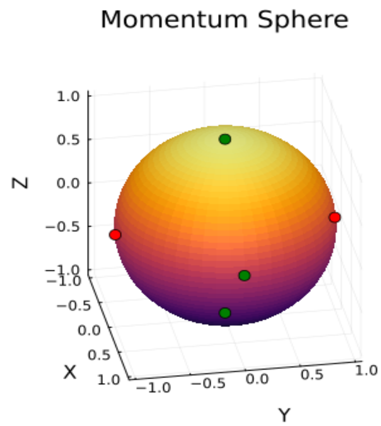


Figure 8: Momentum Sphere View 2

from the angular velocity plot in Figure 5

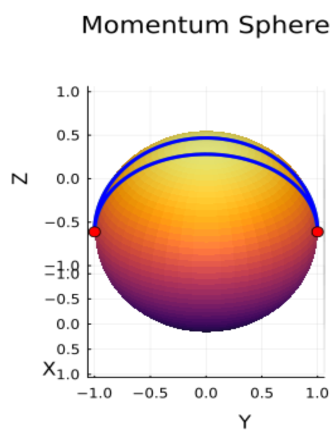


Figure 9: Momentum Trajectory about the Intermediate Axis

Next, we simulated a spin about the minor axis. The initial angular velocity is set to a magnitude of 3 RPM with a disturbance in a random direction with a magnitude of 0.1 $\frac{rad}{s}$. The momentum trajectory about the minor axis is shown in Figure 10. This is a stable equilibria, therefore the momentum trajectory is a periodic orbit about the equilibrium point.

Lastly, we simulated a spin about the major axis with an initial angular velocity magnitude set to 3 RPM with an additive random disturbance with a magnitude of 0.1 $\frac{rad}{s}$. This is also a stable equilibria; therefore, the result is a periodic orbit about the equilibrium point.

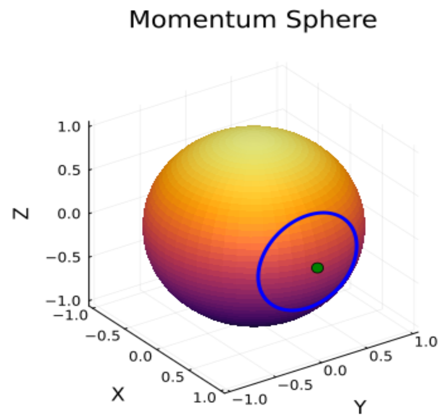


Figure 10: Momentum Trajectory about the Minor Axis

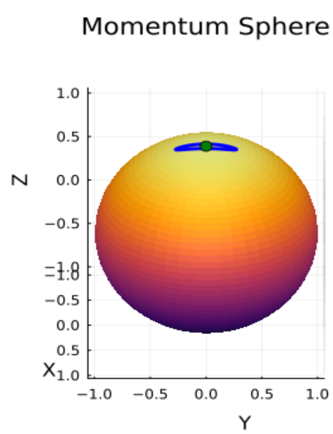


Figure 11: Momentum Trajectory about the Major Axis

Github Repository: `https://github.com/vegaf1/acds-mit-course`

References

[1] N. Alem, T. G. Hendriks, J. L. Alvarellos, P. G. Levinson-Muth, and A. Dono, "Mission design and flight dynamics operations for the starling swarm technology demonstration," in *AAS/AIAA Astrodynamics Specialist Conference*, 2025.

[2] S. Miller, C. Adams, N. Alem, H. Cannon, R. Grashuis, T. Hendriks, S. Hwang, M. Iatauro, C. Pires, J. Kruger *et al.*, "Smallsat 2024-starling cubesat swarm technology demonstration flight results," in *Proceedings of the 38th Annual Small Satellite Conference*, no. SSC24-I-06. Small Satellite Conference, 2024.

[3] T. Stevenson, "Flight results and lessons learned from the starling propulsion system," 2024.

[4] NASA RSDO, "Rapid IV Catalog: BCT CubeSat Marketing Sheet (v6)."